MANIPAL UNIVERSITY
JAIPUR

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School of Basic Science

Department of Computer Applications

Program List

PROGRAMMING & PROBLEM-SOLVING USING C LAB | CA6133 | 1 Credit

Session: July – Nov 2026 | Faculty: Dr. Veerpal Kaur & Dr. Kuntal Gaur | Class: MCA (I Sem)

S. No.	Title of the Program
1.	Write a C program to calculate the area and perimeter of rectangle, square and circle.
2.	Write a C program to convert temperature given in Fahrenheit to Centigrade and Centigrade to Fahrenheit. Hint: C=5/9(F-32)).
3.	Write a C program to perform all the arithmetic operations by taking inputs from command line.
4.	Write a C program to illustrate the use of unary prefix and postfix increment and decrement operators.
5.	Write a C program to perform all the bitwise operations by taking inputs.
6.	Write a C program to interchange values of two variables without using a third variable.
7.	Write a C program to find the greatest number among three variables.
8.	Write a C program to check whether the given number is odd or even.
9.	Write a C program to check whether a number is prime or not.
10.	Write a C program to find the largest and smallest among three entered numbers and display whether the identified largest/smallest number is even or odd.
11.	Write a program to compute grade of students using if else adder. The grades are assigned as followed: **Marks** — **Grade** • 90≤marks≤100 — A+ • 80≤marks< 90 — A • 70≤marks< 80 — B+ • 60≤marks< 70 — B • 50≤marks< 60 — C+ • 40≤marks< 50 — C • 30≤marks< 40 — D marks<30 — F
12.	Write a C program to compute the final grade of student's five subject marks. Take entry of 5 subject marks from user out of 100.
13.	Write a C program to check whether the entered year is a leap year or not.
14.	Write a C program to find the factorial of a number.
15.	Write a C program to find whether a character is consonant or vowel using a switch statement.

S. No.	Title of the Program
16.	Write a C program to print the number of days in a given month and year using a switch case.
17.	Write a C program to determine whether the input character is a capital or small letter, digit, or special symbol.
18.	Write a C program to print the sum of digits of a number using a for loop.
19.	Write a C program to check whether a number is Palindrome or not.
20.	Write a C program to generate the Fibonacci series.
21.	Write a C program to reverse and sum of the digit using a while loop.
22.	Write a C program to display the following patterns: * ***** 1 12345 1 ** **** 12 1234 22 *** *** 123 123 333 **** ** 1234 12 4444 ***** * 12345 1 55555
23.	Write a C program to display series and sum of the series: 1+1/2+1/3+……1/n.
24.	Write a C program to Search an element in an array.
25.	Write a C program to perform the addition of all elements in Array.
26.	Write a C program to find the largest and smallest element in Array.
27.	Write a C program to perform the multiplication of two Matrices.
28.	Write a C program to print the reverse of a string without using predefined string functions.
29.	Write a C program to read a string and check for palindrome without using a string string-related function.
30.	Write a C program to accept a string and count the number of vowels present in this string.
31.	Write a C program to find the highest frequency character in a string.
32.	Write a C program to remove all repeated characters from a given string.
33.	Write a C program to add, subtract, multiply, and divide two integers using user user-defined type function with return type.
34.	Write a C program to calculate the factorial of a number using recursion.
	Write a C program to print the Fibonacci series up to a given number using recursion.
34.	Write a C program to swap two integers using call-by-value and call-by-reference methods of passing arguments to a function.
35.	Write a C program to evaluate the equation $y= x^n$ where n is a positive integer using function call by value.
36.	Write a C program to calculate the sum of numbers from 1 to n using recursion.
37.	Write a C program to print all even and odd numbers in a given interval using recursion.
38.	Write a C program to input and print elements of an array using pointers.
39.	Write a C program to reverse an array in place using pointers.
40.	Write a C program to find the reverse of a string using pointers.

41.	Write a C program to perform PUSH and POP operations on the stack.
42.	Write a C program to create and display a queue.
43.	Write a C program to perform insertion and deletion on the queue.

(Dr. Veerpal Kaur)
(Course Coordinator)